

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: BTech

Branch : Mechanical Engineering

Semester :III

Subject Code & Name: BTMEC303_Y19 Fluid Mechanics

Max Marks: 60

Date: 23-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	Which one of the following is the unit of mass density? a. kg/m ³ b. kg/m ² c. kg/m d. kg/ms	Understanding CO1	1
2	 is phenomenon by which liquid rises into a thin glass tube above or below its general level. a. Surface tension b. Capillarity c. Cohesion d. Adhesion	Understanding CO1	1
3	The point of application of the force of buoyancy on the body is known as... a. centre of gravity b. centre of buoyancy c. metacenter d. none of the above.	Understanding CO2	1
4	If the position of metacenter M remains lower than c.g. of the body, G, the body will remain in a state of a. stable equilibrium b. unstable equilibrium c. neutral equilibrium d. any of the above.	Understanding CO2	1
5	The flow in a pipe whose valve is being opened or closed gradually is an example of... a. steady flow b. unsteady flow c. rotational flow d. compressible flow.	Understanding CO3	1
6	If the Reynolds number is less than 2000, the flow in a pipe is.... a. laminar flow b. turbulent flow c. transition flow d. none of the above.	Understanding CO3	1
7	The hydraulic gradient-line indicates the variation of which of the following: a. Velocity head in flow direction b. Piezometric head in the direction of flow c. Total energy of flow in the direction of flow d. None of the above.	Understanding CO4	1
8	Which of the following is NOT a type of force considered in the Navier-Stokes equation? a. Gravity force b. Pressure force c. Surface tension force d. Viscous force	Understanding CO4	1
9	... studied the laminar flow through a circular tube experimentally. a. Prandtl b. Pascal c. Hagen and Poiseuille d. None of the above.	Understanding CO5	1
10	Darcy-Weisbach equation is used to find loss of head due to : a. sudden enlargement b. sudden contraction c. friction d. none of the above.	Understanding CO5	1

11	In order that results obtained in model studies correctly represent the behavior of the prototype, which of the following similarities must be ensured between the model and the prototype?	Understanding CO6	1
	a. Geometric similarity b. Kinematic similarity c. Dynamic similarity d. All of the above.		
12	A streamlined body is defined as a body about which...	Understanding CO6	1
	a. the drag is zero b. the flow is laminar c. the flow is along streamlines d. the flow separation is suppressed.		
Q.2	Solve the following.		12
A)	Explain following terms a) Mass density b) Specific gravity c) Dynamic viscosity d) Kinematic viscosity e) Pascal's law f) Surface tension	Understanding CO1	6
B)	Calculate the capillary effect in millimeters in a glass tube of 4 mm diameter, when immersed in (i) water and (ii) mercury. The temperature of the liquid is 20°C and the values of surface tension of water and mercury at 20°C in contact with air are 0.0735 N/m and 0.51 N/m respectively. The contact angle for water $\theta = 0^\circ$ and for mercury $\theta = 130^\circ$. Take specific weight of water at 20°C as equal to 9790 N/m ³ .	Apply CO1	6
Q.3	Solve the following.		12
A)	Define the terms: Stable, unstable and neutral equilibrium.	Understanding CO2	6
B)	A circular plate of diameter 1.2 m placed vertically in water in such a way that the centre of the plate is 2.5 m below the free surface of water. Determine: (i) Total pressure on the plate. (ii) Position of centre of pressure.	Apply CO2	6
Q.4	Solve Any Two of the following.		12
A)	Write difference between following. a) Laminar and Turbulent flow b) Steady and Unsteady flow	Understanding CO3	6
B)	State Bernoulli's Theorem, mention the assumptions made. How it is modified while applying in actual practice?	Understanding CO4	6
C)	An oil of sp. Gr. 0.8 is flowing through a venturimeter having inlet diameter 0.2 m and throat diameter 0.1m. The oil mercury differential manometer shows a reading of 250 mm. Calculate the discharge of oil through the horizontal venturimeter, Take $C_D = 0.98$.	Apply CO4	6
Q.5	Solve Any Two of the following.		12
A)	Write a short note on following terms: a) Major energy losses b) Minor energy losses.	Understanding CO5	6
B)	Three pipes of lengths 800m, 600m and 300m and of diameters 400mm, 300mm and 200mm respectively are connected in series. The ends of the compound pipe is connected to two tanks whose water surface levels are maintained at a difference of 15m. Determine the rate of flow of water through the pipes if $f = 0.005$. What will be diameter of a single pipes of length 1700m and $f = 0.005$, which replaces the three pipes.	Apply CO5	6
C)	A town having population of 1, 00,000 is to be supplied with water from a reservoir at 6km distance. It is estimated that one-half of daily supply of 125 liters per head should be delivered within 8	Apply CO5	6

	hours. Calculate the diameter of pipe through which water is supplied, if the head available is 16m. Take $c = 45$ in Chezy's formula.		
Q.6	Solve Any Two of the following.		12
A)	Explain dimensional analysis along with use of dimensional analysis and its advantages.	Understanding CO6	6
B)	Explain methods of preventing the separation of boundary layer.	Understanding CO7	6
C)	Explain drag and lift. Give some practical examples related to these terms.	Understanding CO7	6
*** End ***			